

Introduction

Forward Induction

- Subgame Perfection and Sequential Equilibrium are “backward induction” concepts, i.e., players behavior towards the end of the game determines players behavior earlier in the game.
- Backward induction ideas have been formalized in various notions including dynamic programming and sequential rationality.
- Kohlberg and Mertens (1986) proposed a notion of “forward induction:” a player’s options early in a game may determine (and help predict) the player’s behavior later in the game. Their refinement is termed Stability.

Various expressions of forward induction

- Iterative Elimination of Weakly Dominated Strategies (IEWDS).
- Equilibrium Dominance.
- The Intuitive Criterion, Cho and Kreps (1987).
- Divinity, Banks and Sobel (1987).
- Never a Weak Best Response Criterion (NWBR).
- Various formulations of stability, Kohlberg and Mertens (1986)},
`\textcite{govindan-wilson2009}`.

Example

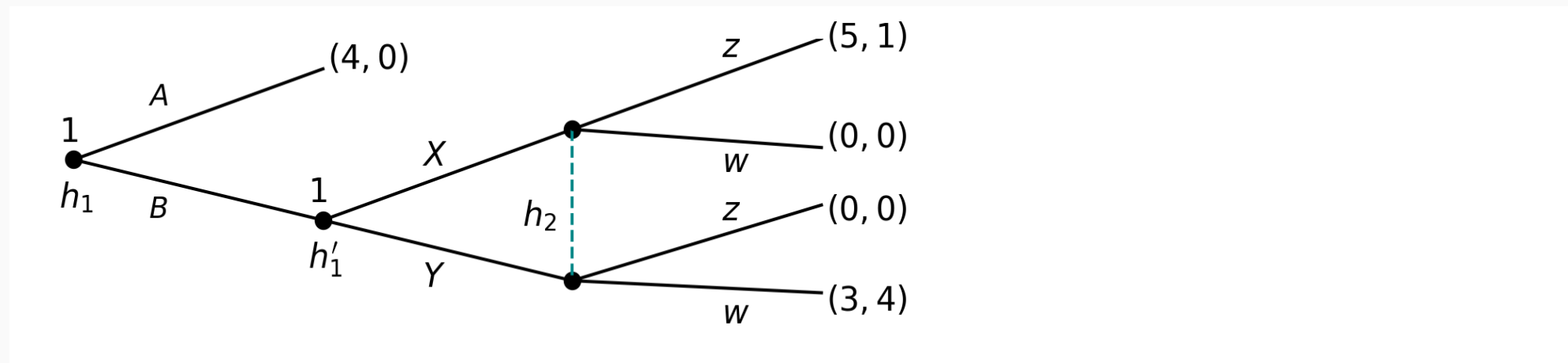


Figure 1

- Three strategy profiles that are part of sequential equilibria. To prove these are SE, beliefs must be provided to construct consistent assessments.
1. $[(A, Y), w]$; 2. $[(A, (.8, .2)), (3/8, 5/8)]$; and 3. $[(B, X), z]$.

Example Figure 1

1. $[(A, Y), w]$

$$\sigma_1(\cdot|h_1) = (\sigma_1(A|h_1), \sigma_1(B|h_1)) = (1, 0)$$

$$\sigma_1(\cdot|h'_1) = (\sigma_1(X|h_1), \sigma_1(Y|h_1)) = (0, 1)$$

$$\sigma_2(\cdot|h_2) = (\sigma_2(Z|h_2), \sigma_2(W|h_2)) = (0, 1)$$

$$\mu_1(h_1|h_1) = 1 \text{ and } \mu_1(h'_1|h'_1) = 1$$

$$\sigma_1(\cdot|h_1) = (\sigma_1(A|h_1), \sigma_1(B|h_1)) = (1, 0)$$

$$\mu_2((B, X)|h_2) = 0 \text{ and } \mu_2((B, Y)|h_2) = 1 \text{ and}$$

$$\sigma_1^n(\cdot|h_1) = \left(1 - \frac{1}{n}, \frac{1}{n}\right)$$

$$\sigma_1^n(\cdot|h_1') = \left(\frac{1}{n}, 1 - \frac{1}{n}\right)$$

$$\sigma_2^n(\cdot|h_2) = \left(\frac{1}{n}, 1 - \frac{1}{n}\right)$$

$$\mu_2^n(\cdot|h_2) = \left(\frac{1}{n}, 1 - \frac{1}{n}\right)$$

because $\mu_2^n((B, X)|h_2) = \frac{1/n^2}{1/n} = 1/n$.

2. $[(A, (.8, .2)), (3/8, 5/8)]$.

$$\begin{array}{ll} \sigma_1 \longrightarrow & (1, 0), (0.8, 0.2) \\ \sigma_2 \longrightarrow & (3/8, 5/8) \end{array}$$

3. $[(B, X), z]$.

$$\begin{array}{ll} \sigma_1 \longrightarrow & (0, 1), (1, 0) \\ \sigma_2 \longrightarrow & (1, 0) \end{array}$$

Forward induction. Intuition

- Forward induction arguments eliminate the first two equilibria.
- Deviations from proposed equilibrium reveal the intentions of the deviating player in the continuation game.
- In the first equilibrium, player 1's equilibrium payoff is 4. Suppose player 1 deviates from this equilibrium and chooses B . What can player 1 expect?

Example

- Iterative elimination of weakly dominated strategies (IEWDS)

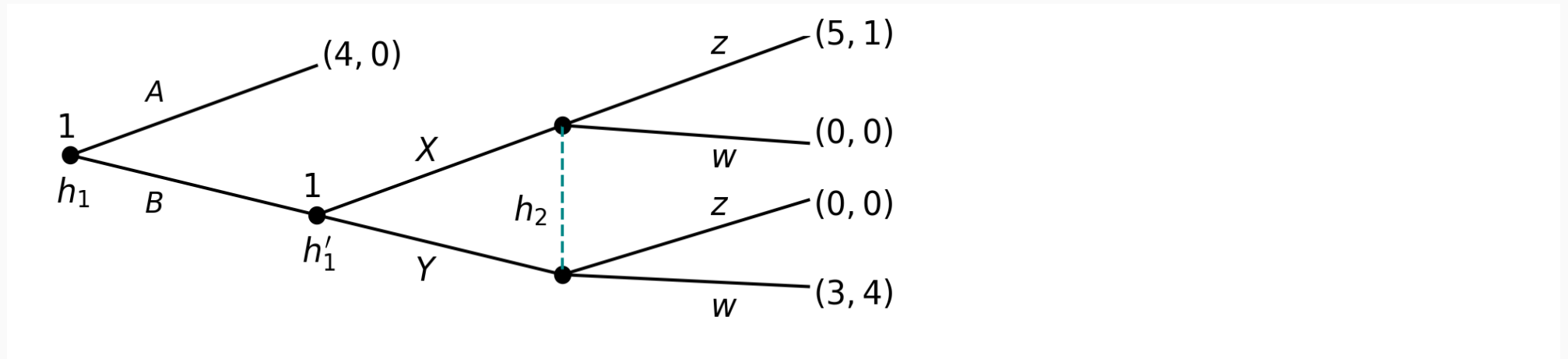


Figure 2

	z	w
A, X	$(4, 0)$	$(4, 0)$
A, Y	$(4, 0)$	$(4, 0)$
B, X	$(5, 1)$	$(0, 0)$
B, Y	$(0, 0)$	$(3, 4)$

- 1's strategy (B, Y) is dominated by (A, Y) .
- Then 2's strategy W becomes weakly dominated by Z .
- The surviving equilibrium is $[(B, X), Z]$.

Example: Equilibrium dominance

- Is $[(AY), w]$ intuitive? Its payoffs are $(4, 0)$.
- Subgame starting at h'_1 has 3 NE:

	(X, Y)	(z, w)	Player 1's Payoff
a.	$(0, 1)$	$(0, 1)$	3
b.	$(0.8, 0.2)$	$(3/8, 5/8)$	$15/8$
c.	$(1, 0)$	$(1, 0)$	5

- Only option c improves player 1's payoff since $5 > 4$.
- Player 1 would only deviate to B (from A) only if she intends to benefit. That suggests that 1 would continue with X .

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- At h_2 , player 2 ought to believe that player 1 chose (B, X) .
 - If 2 reasons as described, 1 should choose (B, X) (i.e., $(0, 1)$ $(1, 0)$).
Player 2 should respond $(1, 0)$.
 - Player 2 infers 1's behavior from the fact that the player deciding at h'_1 is the same as the player deciding at h_1 .
 - This suggests that an analytic formulation must depend on the Normal form and not on the Agent Normal Form.

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- In the example, the intuition behind “forward induction” may be formalized by the *iterative elimination of weakly dominated strategies*.
 - In other games, IEWDS yields a backward induction equilibrium.
 - Still in other games, IEWDS yields either a backward induction equilibrium or a forward induction equilibrium depending on the order of elimination of strategies.
 - Forward induction is based on the principle that deviations should be explained rationally; backward induction on the principle that deviations are mistakes.

Example: forward-backward conflict

- Conflict between forward and backward induction IEWDS. Myerson (1991), pp. 191-196.

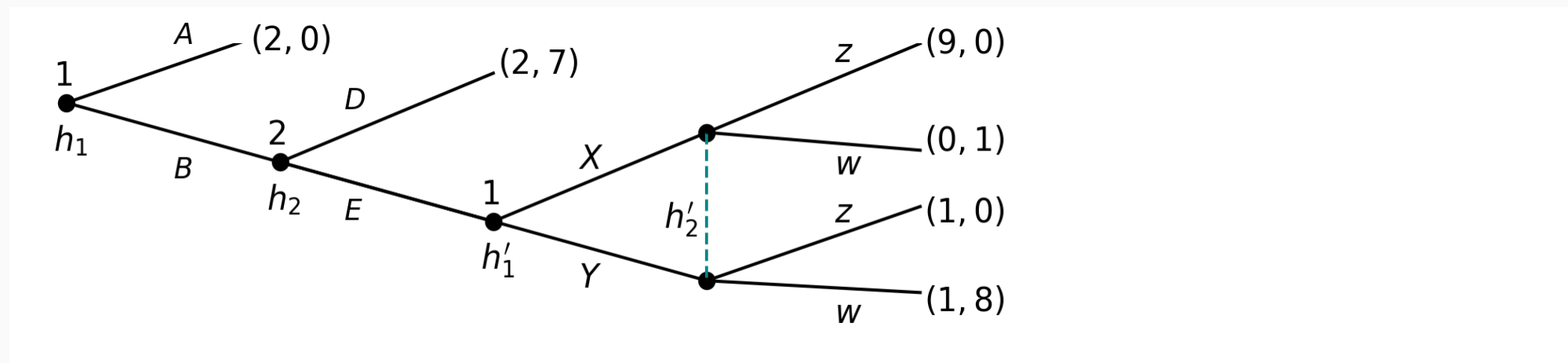


Figure 3

1. Process starts with 2, strategy (E, z) . Only one strategy is eliminated in each iteration:

- 2's strategy (E, z) weakly dominated by all others,
- 1's strategy (B, X) weakly dominated by all others,
 (B, Y) also dominated but it is not eliminated at this stage.
- 2's strategy $(D, \cdot)$ is weakly dominated by (E, w) ,
- OUTCOME: $[(A, \cdot), (E, w)]$. This is the backward induction equilibrium.

2. The process starts with player 1, strategy (B, Y) .

- 1's strategy (B, Y) weakly dominated by $(A, \cdot)$,
- 2's strategy (E, z) weakly dominated by all others,
- 1's strategy (B, X) weakly dominated by $(A, \cdot)$,
- OUTCOMES: $[(A, \cdot)(D, \cdot); (A, \cdot)(E, w)]$.

The forward induction equilibrium is contained in the set of outcomes obtained by IEWDS.

References

- Banks, Jeffrey S., and Joel Sobel. 1987. "Equilibrium Selection in Signaling Games." *Econometrica* 55 (3): 647–61. <https://doi.org/10.2307/1913604>.
- Cho, In-Koo, and David Kreps. 1987. "Signaling Games and Stable Equilibria." *Quarterly Journal of Economics* 102: 179–222.
- Kohlberg, Elon, and Jean-François Mertens. 1986. "On the Strategic Stability of Equilibria." *Econometrica* 54 (5): 1003–37.